CISCO NETWORKING BASICS

Practical Project Report

Small Office Local Area Network (LAN)
Design, Configuration & Troubleshooting using Cisco Packet Tracer

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Project level	Beginner – Practical
Tool	Cisco Packet Tracer
Core topics	LAN, IPv4, subnet mask, gateway, router, switch, ping, troubleshooting

A beginner-friendly, step-by-step lab report with an IP addressing plan, device configuration, verification commands, expected results and troubleshooting guidance.

1. Project Overview

Project title: Small Office LAN Design and Configuration using Cisco Packet Tracer.

Scenario: A small office needs a basic wired network for four employee computers and one internal server. A router will act as the default gateway, while two switches provide wired connectivity.

Objective: Build a simple LAN, assign IPv4 addresses, configure router interface settings, verify end-to-end connectivity and troubleshoot common faults.

Learning outcomes: Understand basic network devices and cabling; create a topology; configure IPv4 settings; use basic Cisco IOS commands; test with ping; and document results.

This is a simulated learning lab. It does not represent a live production network or a Cisco-issued project certification.

2. Network Components

Device	Quantity	Purpose
Cisco router (e.g., 2911 or 1941)	1	Provides the LAN default gateway
Cisco switch (e.g., 2960)	2	Connects wired devices within the LAN
PC-1 to PC-4	4	Employee workstations
Server	1	Internal server for connectivity testing
Copper straight-through cables	As needed	Connect PCs/server to switches and switch to router

Software: Cisco Packet Tracer. Device model names and interface labels can vary by Packet Tracer version. If your router uses a different interface name, select the matching Ethernet/GigabitEthernet port shown in its Config or CLI tab.

3. Proposed Topology

Build the topology in this logical order:

```
PC-1 ----\
PC-2 ----- SW1 -------- SW2 ----- PC-3
           |             |        PC-4
           |             |        Server
           +---- Router G0/0
```

Suggested physical links: PC-1 and PC-2 to SW1; SW1 to SW2; PC-3, PC-4 and Server to SW2; and SW1 to Router G0/0. Use copper straight-through cables. Auto-Connect in Packet Tracer may also select a suitable cable.

Note: The router connects to SW1. All end devices are in one IPv4 subnet, so no inter-VLAN routing is required for this beginner lab.

4. IP Addressing Plan

Device	Interface	IPv4 address	Subnet mask	Default gateway
Router	G0/0	192.168.1.1	255.255.255.0	—
PC-1	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC-2	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC-3	FastEthernet0	192.168.1.4	255.255.255.0	192.168.1.1
PC-4	FastEthernet0	192.168.1.5	255.255.255.0	192.168.1.1
Server	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.1

Subnet: 192.168.1.0/24. **Usable host range:** 192.168.1.1–192.168.1.254. **Broadcast address:** 192.168.1.255. Each device must have a unique IP address.

DNS is not required for basic IP-to-IP ping testing. If you later enable a DNS service, enter that DNS server address on the clients.

5. Build the Topology in Packet Tracer

Step 1. Open Cisco Packet Tracer and create a new project.

Step 2. From Network Devices, place one router and two 2960 switches on the workspace.

Step 3. From End Devices, place four PCs and one Server.

Step 4. Connect PC-1 and PC-2 to SW1; connect SW1 to SW2; connect PC-3, PC-4 and Server to SW2; connect SW1 to the router G0/0 port.

Step 5. Wait for link indicators to turn green. An amber indicator may appear briefly while a switch port initializes.

Step 6. Save the file with a clear name, for example: Small_Office_LAN_Bipul_Dubey.pkt.

6. Configure the Router

Click the router, open the CLI tab and press Enter. If the initial configuration dialog appears, type **no** and press Enter. Enter the following commands:

```
enable
configure terminal
hostname R1
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
end
copy running-config startup-config
```

If asked for the destination filename, press Enter to accept the default. If the router interface is named GigabitEthernet0/0/0 or FastEthernet0/0 in your model, use that exact interface name instead.

Explanation: enable enters privileged EXEC mode; configure terminal enters global configuration mode; hostname labels the router; interface selects the LAN port; ip address assigns the gateway address; no shutdown enables the interface; and copy saves the running configuration.

7. Configure the Switches (Basic)

For each switch, open CLI and enter a hostname. A basic Layer 2 switch can forward traffic without a management IP address.

```
enable
configure terminal
hostname SW1
end
copy running-config startup-config
```

On the second switch, use **hostname SW2** instead. No VLAN or trunk configuration is needed for this single-subnet beginner topology. Leave the switch ports in their default access VLAN.

8. Configure PCs and Server

For each PC and the server:

Step 1. Click the device and open Desktop.

Step 2. Select IP Configuration.

Step 3. Choose Static.

Step 4. Enter the IP address, subnet mask and default gateway exactly as shown in the addressing table.

Step 5. Close the window. Repeat for every end device, using a different IP address for each.

Example for PC-1: IP address 192.168.1.2; subnet mask 255.255.255.0; default gateway 192.168.1.1. On the server, use 192.168.1.10 with the same mask and gateway.

9. Verify Connectivity

On PC-1, open Desktop → Command Prompt and run:

```
ipconfig
ping 192.168.1.1
ping 192.168.1.3
ping 192.168.1.10
```

Repeat suitable ping tests from other PCs. A successful test displays replies from the destination. The first ping may time out while ARP resolves a MAC address; run it again before diagnosing a fault.

On the router CLI, verify the interface and configuration:

```
show ip interface brief
show running-config
show interfaces gigabitEthernet 0/0
```

Expected router status: the configured LAN interface should show IP address 192.168.1.1 and status/protocol up/up when the cable and switch connection are working.

10. Expected Results

Test	Expected result
PC-1 → Router (192.168.1.1)	Ping replies received
PC-1 → PC-2 (192.168.1.3)	Ping replies received
PC-1 → Server (192.168.1.10)	Ping replies received
PC-3 → PC-4 (192.168.1.5)	Ping replies received
Router G0/0	Interface up/up after link is active

Record your own Packet Tracer test results and insert screenshots in the evidence section. Do not claim a test passed unless you ran it in your own simulation.

11. Troubleshooting Guide

Problem	Checks and corrective action
Ping says Destination host unreachable	Check IP, subnet mask, cable, switch link and router interface. Confirm devices are in the same subnet.
Request timed out	Check destination device is powered on, IP settings are correct, and links are active. Retry once after ARP
Router interface is administratively down	Enter the correct interface configuration mode and run no shutdown.
Duplicate IP address	Give each PC/server a unique address from the addressing plan.
Wrong default gateway	Set each end device's gateway to 192.168.1.1.
No link light	Check cable type, selected ports, device power and interface status.

Useful commands:

```
show ip interface brief
show running-config
show interfaces
show mac address-table
```

```
ipconfig
ping 192.168.1.1
```

12. Security and Good Practice

For a learning lab, use unique device names, document IP addresses, save configurations, and avoid sharing passwords. In a real office, also apply approved device access controls, secure management access (SSH rather than Telnet), strong credentials, backups, updates, and network segmentation according to the organization's policy. Do not expose management services unnecessarily.

13. Conclusion

This project demonstrates the basic process of designing and configuring a small wired LAN in Cisco Packet Tracer. It covers device placement, cabling, IPv4 addressing, router gateway configuration, basic switch setup, connectivity testing and troubleshooting. Completing the lab provides practical experience with foundational networking concepts.

Future improvements: Add DHCP to assign addresses automatically, create separate VLANs for departments, configure inter-VLAN routing, add an access-control list, and test a second subnet.

14. Practical Evidence Checklist

■ Screenshot 1 – Completed Packet Tracer topology

■ Screenshot 2 – Router CLI showing show ip interface brief

■ Screenshot 3 – PC-1 IP Configuration

■ Screenshot 4 – Successful ping to default gateway

■ Screenshot 5 – Successful ping to another PC and the server

■ Saved .pkt Packet Tracer project file

15. Viva / Interview Questions

Q. What is a LAN?
Ans. A Local Area Network connects devices within a limited area such as an office or building.

Q. What is a router?
Ans. A Layer 3 device that forwards packets between different IP networks.

Q. What is a switch?
Ans. A device that forwards Ethernet frames within a LAN, typically using MAC addresses.

Q. What is an IP address?
Ans. A logical address used to identify a device on an IP network.

Q. What is a subnet mask?
Ans. It identifies the network and host portions of an IPv4 address.

Q. What is a default gateway?
Ans. The router address a host uses to reach destinations outside its local subnet.

Q. What does ping test?
Ans. It tests IP reachability using ICMP Echo messages and replies.

Q. What does no shutdown do?
Ans. It administratively enables a router interface.

Q. What is the purpose of DNS?
Ans. DNS translates domain names into IP addresses.

Q. What is DHCP?
Ans. DHCP automatically provides IP configuration to clients.

Appendix: Project Details for Resume

Small Office Network Design using Cisco Packet Tracer

Designed and simulated a small office LAN using a Cisco router, two switches, four PCs and a server. Configured IPv4 addressing and default gateways, enabled the router interface, practiced basic Cisco IOS commands, verified connectivity using ping, and performed basic network troubleshooting.

Important: Add this project to your resume after you have personally built and tested it. Be prepared to explain the topology, addressing plan and commands during an interview.

Prepared as a learning project report for Cisco Networking Basics. Device interfaces and Packet Tracer menus may differ by software/device version.